



10-152-117 Python Programming Foundations

Python Programming Fall 2026 Syllabus

Cash Myers

SOUTHWEST TECH

INSTRUCTOR

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Office Location: Remote (via zoom)

Office Hours:

Wed mornings

CLASS INFORMATION

Section Number:

Term: Fall Year: 2026 Start Date: 8/17/2026 End Date: 10/9/2026

Meeting Location: Remote

Delivery Mode: Online

COURSE INFORMATION

Alternate Title: Python Programming

Learners explore Python programming by designing and implementing solutions to real-world problems. Students develop foundational programming skills using variables, data types, conditionals, loops, functions, and data structures. The course emphasizes problem-solving, debugging, and code comprehension while introducing the use of AI-assisted tools to support development. Learners practice building, evaluating, and refining programs, both independently and with AI assistance. By the end of the course, students are able to create, understand, and explain basic Python programs and apply structured thinking to solve problems.

Instructional Level: A.A.S. - Associate in Applied Science

Credits: 2

Total Hours: 54

Types of Instruction:

- 18: (A) Lecture
- 36: (B) Lab

TEXTBOOKS

Learn Python Programming: A comprehensive, up-to-date, and definitive guide to learning Python - 4th ed. Edition - Fabrizio Romano/Heinrich Kruger

The Missing Tools: How Developers Learn to Think Quickly - 1st Edition - Cash Myers

Python Programming Language: a QuickStudy Laminated Reference Guide Pamphlet – Robin Nixon – (Optional)

SUPPLIES

- Github account for assignment submission
- GitHub Copilot account for AI support
- Laptops assigned from Charger 360 program

GRADING INFORMATION

PYTHON PROGRAMMING GRADING SUMMARY

Each assignment may be assessed in up to **12 rubric categories**. Each category is scored from **0 to 4 points**, so a fully assessed assignment is worth up to **48 points**.

Your course grade includes **16 assignments** organized into two categories:

Category	Assignments	Weight in final grade
Lab	A1-A14 (14 assign)	77.78%
Capstone Submission	A15-A16 (2 assign)	22.22%

The Capstone Build and AI Use Justification and Final Presentation assignments each count twice as much as a regular assignment in your final grade.

Final Letter Grade Scale

Final grades are calculated from your weighted course percentage.

Letter grade	Final grade percentage
A	90-100%
B	80-89%
C	70-79%
D	60-69%
F	0-59%

AI - FULL USE OF AI IN COURSE

Southwest Tech encourages students to explore generative artificial intelligence (GenAI) as a valuable tool for learning and creativity. In this course, GenAI use is permitted and encouraged, provided it supports your learning and adheres to high standards of academic honesty. Clearly acknowledge any significant use of GenAI in your work, ensuring all submissions reflect your understanding and critical evaluation. The misuse of GenAI that misrepresents your skills or knowledge violates academic integrity. For more details, refer to the Student Code of Conduct or contact your instructor.

AI - LIMITED USE OF AI IN COURSE

Southwest Tech encourages students to explore generative artificial intelligence (GenAI) as a tool for learning and productivity. In this course, the use of GenAI is permitted only with prior approval or instruction from the instructor. GenAI may assist with brainstorming, research, or drafting, but its use must align with the instructor's guidelines and should not replace your effort or understanding. All GenAI usage must be clearly acknowledged in your work and comply with the college's academic honesty standards. The misuse of GenAI that misrepresents your skills, or knowledge is a violation of academic integrity. For more details, refer to the Student Code of Conduct or contact your instructor.

ONLINE - ATTENDANCE

In an online course, you are not required to show-up in at a particular location and at a particular time. However, you are expected to show-up online.

This course is designed to be an active and interactive course in which you contribute and respond to the contributions of others. You should plan to log on and participate in the course at least 3-5 times each week. I will measure this by recording the date of each entry you make to the system--a response to an assignment, an entry into the threaded discussion, submission of an assignment, or participation in group work. (Though you are welcome to participate in the CyberCafe as frequently as you would like, this does not qualify as "attending and participating" in the course.)

Please do not "drop out" for a period of time and then expect to "drop back in." If your business/personal obligations or illness require you to be absent for a week, please contact me and arrange to make up the work you will miss. Since you can access this online course from anywhere in the world, there should be little likelihood "absence."

ONLINE - COMMUNICATION

In a cyber community, you present yourself and learn about others through written words. You don't need to be a prize-winning author or poet to successfully communicate in an online community, however you do want to present yourself in a positive light and to communicate your thoughts and ideas effectively.

The following guidelines will help you ensure that you are properly understood, get your points across effectively, avoid getting anybody annoyed, and avoid looking like a "beginner" on the net.

1. Format your posting so that it is easy to read. Use short paragraphs separated by blank lines. Don't write everything in uppercase (capital) letters. It is more difficult to read and, even worse, in an online environment it means you are SHOUTING.
2. Be brief. Plan your messages ahead so that you don't ramble.
3. Be clear. Don't use abbreviations or acronyms that others may not understand. Read your messages over before sending them.
4. Check your spelling. People will not take you seriously, no matter how brilliant your ideas, if your writing is full of misspellings. Use your computer's Spell Check features; then read over what you have written to catch errors that Spell Check misses.

5. When you are interacting with others online, remember that things may "sound" harsh or less friendly when the reader cannot see your smile or the twinkle in your eye. Read your messages over to be sure they are diplomatic and polite.

ONLINE - CORE ABILITIES

Although the primary setting for learning in this course is online, you are a member of a learning community just as you would be in a classroom course. As members of an online learning community, we share responsibility for creating and maintaining an environment that communicates mutual respect, supports learning for all members, and provides opportunities for all members of the community to learn from one another. Throughout this learning experience we will strive to build the following online learning core abilities:

Learn effectively

You will know you are practicing effective online learning habits:

- o you actively engage in the online learning community
- o you use the syllabus, learning plans, performance assessment tasks, and learning materials to guide learning
- o you use online learning tools such as Profiles, Orientation, Learning Plans, Discussion, Chat, In/Out Box, WorkSpace, GradeBook, Help Desk
- o you take responsibility for self as a learner
- o you use resources that are provided and find additional resources to meet learning needs
- o you produce evidence of learning that meets the performance expectations

Work cooperatively

You will know you are effectively contributing to the online learning community when:

- o you complete assigned tasks for team/group work
- o you use collaborative strategies to complete tasks
- o you exchange information, ideas, and opinions in group and/or class discussions
- o you actively provide feedback through Peer Review process

Act responsibly

You will know you are taking responsibility for your online learning when:

- o you complete assigned tasks according to prescribed deadlines
- o you complete assigned tasks according to prescribed criteria
- o you are an active participant in your cyber community
- o you observe the rules of netiquette

Think critically and creatively

You will know you are thinking critically and creatively when:

- o you respect other points of view
- o you apply the principles and strategies of purposeful, organized thinking to problem solving and decision making
- o you distinguish between fact and opinion
- o your contributions to online discussions show original thought
- o you synthesize information from a variety of sources

ONLINE - GROUP WORK

Online Group Work

In this course you will be required to collaborate on projects with a Work Group. Your instructor will assign you to a work group of 3-5 people. Your work group assignment will be posted in the Announcements or Discussion under work groups. You will use the same work group for the entire course.

1. Your work group will be assigned a name and number. Your group will share a WorkSpace where you can post and review each other's work, and a Work Group Discussion Forum for you to share thoughts and ideas. Only members of your group (and your instructor) can access what's posted in these areas.
2. Take time to review the Profiles of other members of your work group. (This is a good time to make sure your profile is completely filled out.) Email a greeting and letter of introduction to your work group members. Indicate the best way and time to reach you.
3. When you are assigned to work on a project collaboratively, post any documents you are sharing to your group's workspace. Use your group's discussion area to post comments. Use email to contact each other as well. When necessary, plan to meet in the course Chat at the same time for synchronous discussion. You may find that you want to exchange your work for peer feedback even when it is not part of the assignment. Your Group WorkSpace is yours to use!
4. After your group has completed an assignment, make sure you send your assignment to me as an attachment to an email or submit it using the InBox.

For groups to work together effectively, it is essential that you respond to your email and discussion postings from group members within 48 hours (preferably 24 hours). If you have difficulty working with a member of your group, please contact me.

ONLINE - INSTRUCTOR ROLE AND RESPONSIBILITIES

As your instructor, I am responsible for providing an environment in which an opportunity for learning exists. I will work with you and assist you in your quest for understanding. I cannot make you learn anything. As a resource person and facilitator, I will organize the course, schedule learning activities, and evaluate the short-run "products" of your learning process. Recognizing that even asynchronous online communication is time sensitive, I will monitor threaded discussions and respond to queries within 48 hours of the time they were posted. Since the Writing and Thinking Skills and Individual Reflections are significant pieces of writing, it may take a little longer to assess them. I will be giving your work, as well as that of your fellow learners, careful consideration.

ONLINE - LEARNER ROLE AND RESPONSIBILITIES

You are an adult learner and as such you are responsible for your own learning. No one else can be a "stand in" for you in the learning process. You will be held accountable for all assigned activities. You matter and what you do does make a difference. You will have an opportunity to share your unique ideas and experiences with your student peers and instructor. The form and content of your participation will determine the level of achievement, satisfaction, and enjoyment that you experience. Because others are depending on you to keep the course moving, you have an obligation to meet deadlines for completing assignments and postings.

ONLINE - PARTICIPATION

This workshop is designed to provide a forum in which you, as a practicing professional, can work together with others to build your [name skills here] skills. We've designed many of the learning activities to be hands-on and collaborative. Your participation in all discussion forums is important to your professional growth and to the learning of the other participants. Participation in discussion is defined as:

Sharing your ideas and examples

Responding to each question with at least a full paragraph answer

ACADEMIC HONESTY 1

Students are expected to do their own work unless advised that collaboration is acceptable. This means that you may use facts from other sources if you re-write them in your own words. Anytime you quote directly from another source or paraphrase substantially, you must cite the source you used. When you take a test, you are expected to keep your eyes on your own paper and protect your test paper from being copied by a classmate.

Failure to use proper citation procedure is considered plagiarism. Plagiarism will result in a grade of "0" if it is flagrant and/or deliberate. Copying from another person's paper or test is academic dishonesty and will result in a grade of "0" for that assignment. In addition, you will be referred to student services for discipline based on college policy.

CORE ABILITIES

Act Professionally

Criteria

- Take responsibility for one's own actions.
- Conform to the technical standards of a profession.
- Practice morally responsible behavior.
- Take responsibility for staying up-to-date
- Maintain confidentiality.
- Exhibit respect for people and property.
- Exhibit a sense of global awareness.
- Display appearance appropriate for work environment.

Communicate Clearly

Criteria

- Speak clearly so others can understand.
- Write with clarity so that others can understand.
- Select appropriate means to convey a message.
- Ask questions for clarification.
- Participate in discussions and group presentations.
- Interpret nonverbal communications.
- Use active listening skills.
- Apply standards of spelling, English, grammar, and punctuation.

Solve Problems

Criteria

- Use critical, creative thinking skills.
- Apply problem solving steps.
- Demonstrate open-mindedness.
- Evaluate alternatives to choosing a solution.
- Implement solutions appropriately.
- Use research to solve problems.

- Use appropriate mathematical calculations.

Value Learning

Criteria

- Assume responsibility for lifelong learning.
- Identify own learning needs.
- Apply effective learning processes.
- Help other to learn effectively.

Work Cooperatively

Criteria

- Demonstrate respect for self and others.
- Contribute to a group activity with ideas, suggestions, and effort.
- Complete own share of tasks necessary to finish a group project.
- Maintain a safe and healthy work environment for self/group.
- Set goals, standards, limits, for self/group.
- Display effective interpersonal skills.
- Resolve conflicts in a constructive manner.
- Seek help when needed.
- Demonstrate ability to work with a diverse population.

Work Productively

Criteria

- Attend regularly and on time.
- Exhibit organizational skills.
- Locate resources for problem solving.
- Display productive work ethic.
- Maintain necessary knowledge and skills.
- Use effective, efficient processes.
- Show self-direction in starting tasks.
- Demonstrate reliability.
- Follow directions.
- Complete required tasks on time.

COURSE COMPETENCIES

1. Apply Python fundamentals to construct working programs

Assessment Strategies

- Lab-based programming assignments
- Project assignments

Criteria

- Write Python statements using correct syntax
- Use variables, values, expressions, and basic data types appropriately
- Apply input and output operations in simple programs
- Interpret and correct basic syntax or type-related issues
- Produce readable code that runs as intended

Learning Objectives

- Identify Python syntax, values, variables, and expressions
- Explain how basic data types are used in Python programs
- Describe how input and output support program interaction

2. Design program logic using control flow and sequencing

Assessment Strategies

- Lab-based programming assignments
- Project assignments
- Logic tracing activities

Criteria

- Break problems into step-by-step program logic
- Use conditionals, loops, and Boolean logic appropriately
- Sequence instructions so program behavior matches the task
- Identify inputs, outputs, constraints, and expected behavior
- Trace program flow across expected cases

Learning Objectives

- Define program logic and control flow
- Explain the role of conditionals, loops, and sequencing
- Describe how problem requirements become program behavior

3. Organize Python code using functions, collections, and readable structure

Assessment Strategies

- Lab-based programming assignments
- Project assignments
- Code review or explanation activities

Criteria

- Define and use functions to organize program behavior
- Use parameters, return values, and collections appropriately
- Choose structures that make code easier to read and maintain
- Reduce unnecessary repetition in program code
- Recognize simple class-based or application structures when introduced

Learning Objectives

- Explain how functions support code organization

- Describe how lists, dictionaries, and other collections organize data
- Identify characteristics of readable and maintainable code

4. Handle data using collections, files, structured formats, and integrations

Assessment Strategies

- Lab-based programming assignments
- Project assignments
- Data-focused programming tasks

Criteria

- Store and retrieve data using appropriate Python structures
- Read from and write to files when required
- Work with structured formats such as CSV or JSON at an introductory level
- Interpret data from simple external or simulated sources
- Select data representations that support the program task

Learning Objectives

- Identify common Python data structures and file patterns
- Explain how structured data can be read, stored, and represented
- Describe how data handling supports larger program behavior

5. Test, debug, and validate Python programs

Assessment Strategies

- Lab-based programming assignments
- Project assignments
- Debugging exercises
- Program explanation checks

Criteria

- Identify syntax, runtime, and logic errors
- Use sample inputs to test program behavior
- Compare expected results with actual results
- Repair program issues and validate the fix
- Explain why a corrected solution works

Learning Objectives

- Define debugging, testing, and validation
- Identify common Python programming errors
- Explain how test cases support program correctness

6. Use AI/RBA-assisted development responsibly and transparently

Assessment Strategies

- Project assignments
- AI-use justification activities
- Prompting and revision exercises

- Final project documentation

Criteria

- Use AI tools within assignment boundaries
- Frame prompts that describe intended program behavior
- Evaluate AI-generated code before using it
- Adapt, reject, or revise AI output based on requirements
- Explain the human decisions made during AI-assisted development

Learning Objectives

- Describe how AI tools can support programming work
- Identify appropriate and inappropriate uses of AI assistance
- Explain how responsible AI use requires validation and accountability

7. Solve programming problems using structured strategies

Assessment Strategies

- Lab-based programming assignments
- Troubleshooting activities
- Project assignments
- Instructor observation

Criteria

- Identify the source of a programming or workflow problem
- Apply problem-solving steps before requesting direct answers
- Use resources, examples, documentation, or targeted support to recover
- Compare possible solutions before selecting an approach
- Persist through revision when an initial solution does not work

Learning Objectives

- Describe structured problem-solving strategies for programming tasks
- Explain how troubleshooting supports independent recovery
- Identify resources that can support technical problem solving

8. Communicate programming logic and development decisions clearly

Assessment Strategies

- Code explanation activities
- Project documentation
- AI-use justifications
- Presentations or discussion

Criteria

- Explain what their code does in clear language
- Describe the logic and decisions behind a solution
- Ask clarifying questions when requirements are unclear
- Document AI use, sources, or design choices when required

- Communicate project outcomes with appropriate technical vocabulary

Learning Objectives

- Explain the role of communication in programming work
- Identify ways to describe code behavior and design decisions
- Describe how clear documentation supports technical accountability

9. Work productively through lab time, iteration, and assignment completion

Assessment Strategies

- Lab-based programming assignments
- Project milestones
- Instructor observation
- Assignment submission patterns

Criteria

- Use lab time to make visible progress on programming tasks
- Organize files, notes, and assignment materials
- Follow directions and assignment requirements
- Complete required tasks by stated deadlines
- Demonstrate self-direction when beginning or continuing work

Learning Objectives

- Identify productive work habits for programming labs
- Explain how organization supports assignment completion
- Describe how iteration contributes to stronger technical work

10. Apply feedback, resources, and revision to improve learning

Assessment Strategies

- Revised assignments
- Reflection activities
- Project checkpoints
- Instructor observation

Criteria

- Use instructor feedback to improve program quality
- Locate and apply appropriate learning resources
- Revise code, explanations, or project decisions based on evidence
- Ask useful questions that support deeper understanding
- Demonstrate growth across repeated programming tasks

Learning Objectives

- Explain how feedback supports learning in programming
- Identify resources that can improve technical understanding
- Describe how revision strengthens both code and reasoning

11. Collaborate respectfully in programming and technical learning activities

Assessment Strategies

- Pair or group activities
- Peer support activities
- Discussion or presentation
- Instructor observation

Criteria

- Participate respectfully in technical learning activities
- Contribute ideas, effort, or support during collaborative work
- Seek help appropriately when blocked
- Support peers without replacing their learning or work
- Resolve confusion or disagreement constructively

Learning Objectives

- Describe expectations for respectful technical collaboration
- Explain how peer support can strengthen learning
- Identify appropriate boundaries for collaboration in programming work

12. Act professionally in technical coursework and tool use

Assessment Strategies

- Lab participation
- Project assignments
- AI-use documentation
- Instructor observation
- Final project presentation or submission

Criteria

- Take responsibility for their work, conduct, and tool use
- Follow course, assignment, and technical standards
- Use AI, code examples, and external resources ethically
- Respect people, property, data, and learning environments
- Demonstrate readiness, reliability, and accountability in technical tasks

Learning Objectives

- Explain professional expectations in programming coursework
- Identify ethical considerations in AI-assisted technical work
- Describe how accountability supports trust in software development

DISABILITY STATEMENT

At Southwest Wisconsin Technical College, we're committed to creating an inclusive and supportive learning environment for all students. If you have a disability, or think you have a disability that could affect your performance in this class, we encourage you to reach out to Disability Services for support as soon as possible. To start a conversation, please call (608) 822-2632, email disabilityservices@swtc.edu, or visit www.swtc.edu/disabilityservices. Note that accommodations are tailored to each class and term and become effective once the registration process with Disability

Services is complete. All services are confidential and designed to provide you access. If you have already been approved for accommodations through Disability Services, please meet with your instructor to make a plan to implement your accommodations.

FINANCIAL AID POLICY

All students should be knowledgeable about Financial Aid policies and responsibilities found on SWTC's Financial Aid Website. Even students who are not currently receiving financial aid need to review this information.

<https://www.swtc.edu/admissions/paying-for-college/financial-aid>

STUDENT CODE OF CONDUCT

Southwest Tech promotes a learning-centered environment dedicated to the advancement of personal growth and knowledge. The College believes every student has the right to pursue an education in a community that values respect, integrity, and inclusivity. To this end, the College prohibits conduct on its campus and in its student housing, which impedes learning, disrupts a peaceful campus environment, and does not promote a cohesive culture based on professionalism, fairness, and trust.

[Student Code of Conduct](#)

TECHNICAL SKILL ATTAINMENT (TSA)

As a program graduate you will have gained the industry-recognized technical skills needed to enter employment in your field. You will have an opportunity to demonstrate these skills through a variety of methods, as determined by your instructors. The details of how you will demonstrate your skills and the level of performance needed for success will be clearly communicated to you within each associated course.

TITLE IX STATEMENT (2024)

At Southwest Tech, we are committed to supporting all students, including those who are pregnant or experiencing related conditions. As part of Title IX, students who need changes to our policies, practices, or procedures due to their pregnancy or related condition(s) will be provided with reasonable modifications to protect their health and educational access. Students may take a voluntary leave of absence, which will be excused, for, at minimum, the period of time deemed medically necessary by a student's licensed healthcare provider and students will be given the opportunity to make up missed work.

For assistance with modifications, time off, lactation breaks/space, or other related needs, please visit the [Pregnancy Support page](#) of the Southwest Tech website and complete the Pregnancy Adjustment Request Form. For discrimination concerns or issues, contact the Title IX staff listed on the [Title IX](#) page.

WITHDRAWING FROM THIS COURSE

The last day to withdraw from this course is to **80%** completion. Please refer to the [Student Handbook](#) for complete details.

YOU BELONG HERE!

Southwest Tech is committed to providing a welcoming learning environment and a sense of community where all students can experience success. We empower and inspire all members of the Southwest Tech community to embrace

differences, defend human dignity, and respect the richness of values and ideas that each person brings to the classroom and the college.